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SECP3213-01 KEPINTARAN PERNIAGAAN (BUSINESS INTELLIGENCE)

SECTION 1

PROJECT TITLE:

**IMPACT ANALYSIS OF GEOGRAPHIC, ECONOMIC, AND SEASONAL
DETERMINANTS ON RETAIL SALES PERFORMANCE**

PHASE 2

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1.0 Workflows Explanation

1.1 Data Input

This project started with data input from the Retail Data Analytic Dataset. Three CSV files are included in the dataset: features data set.csv, sales data set.csv, and stores data set.csv. The **Input Data Tool** was used to input each CSV file from the local storage into the Alteryx Designer workflow. The following Figure 1 is the full workflow for the data input process.

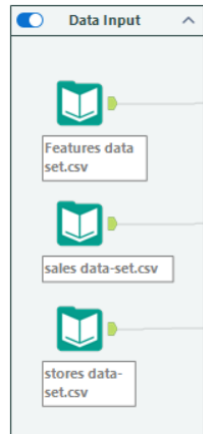


Figure 1: Data Input Workflow

1.2 Data Cleaning

Following data input, the datasets were thoroughly cleaned to ensure data integrity, eliminate duplicates, and resolve missing values across key variables.

1. Feature dataset (First Line):

- a. **DateTime Tool:** The Date column was converted from String to Date/Time in the format of dd/MM/yyyy and a new date column named Date_New was added.
- b. **Select Tool:** Used to rename the new date column to Date and changed the data type for Temperature, Fuel_Price, Markdown1, Markdown2, Markdown3, Markdown4, Markdown5, CPI, and Unemployment to **Double** data type and IsHoliday to **Bool** data type. All columns are selected except the old Date column.
- c. **Data Cleansing Tool:** This tool was used to replace null values in MarkDown 1-5 columns with 0.
- d. **Imputation Tool:** The mean value of each column is used to replace the null value in the CPI and Unemployment columns.

2. Sales dataset (Second Line):

- a. **DateTime Tool:** The Date column was converted from String to Date/Time in the format of dd/MM/yyyy and a new date column named Date_New was added.
- b. **Select Tool:** The new date column was renamed to Date using this tool, and the data types for Store and Dept, Weekly_Sales, and IsHoliday were changed to Int64, Float, and Bool, respectively. All columns are selected except the old Date column.
- c. **Unique Tool:** This tool was used to check duplicate data from the combination value of 3 columns: Store, Dept, and Date.

3. Store dataset (Last Line):

- a. **Select Tool:** This tool was used to change the data type of the Store and Size columns to **Int64**. All columns are selected.

The following Figure 2 is the full workflow for the data cleaning process.

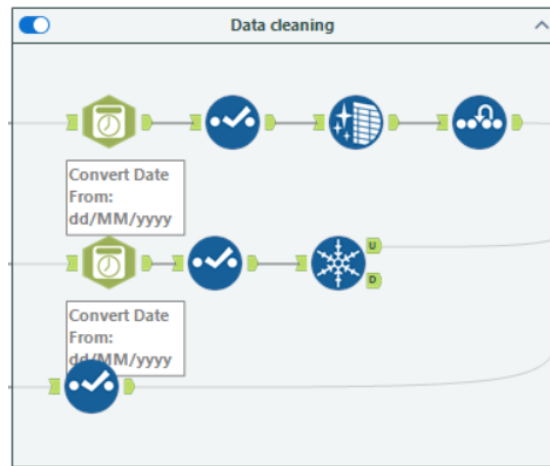


Figure 2: Data Cleaning Workflow

1.3 Data Transformation and Integration

To create a unified business intelligence data model, the three distinct cleaned datasets which are Sales, Stores, and Features were systematically integrated.

1. **Join Tool First:** Combines the **Sales** dataset and the **Features** dataset. A composite key matching on both Store and Date was applied because external environmental factors like CPI, Temperature and Fuel Prices fluctuate weekly alongside sales performance. Unnecessary duplicate fields were deselected in the configuration matrix to optimize memory usage.
2. **Join Tool Second:** Integrates the combined sales-features matrix with the **Stores** dataset. The datasets were merged on the common field Store to append static metadata like Type and Size to every historical weekly transaction record.

Following integration, the dataset underwent several downstream transformations to extract macro-level business insights and create new analytical features.

1. **Summarize Tool (Summarize 10):** Used as the foundational step for store tiering. It groups data strictly by individual Store and calculates the total lifetime revenue using **Sum** on Weekly_Sales resulting in a new field: Sum_Weekly_Sales.
2. **Formula Tool:** Constructs a custom segmentation feature based on revenue performance. A conditional structure categorizes stores into three performance tiers, “Low Volume”, “Medium Volume” and “High Volume” based on natural data thresholds.
3. **Join Tool Third:** Re-integrates the newly created Store_Performance_Tier back into the main workflow stream. By joining the primary data branch with this segmentation

sub-branch on the common Store key, every historical line item is successfully tagged with its respective store's macro performance classification.

The workflow for the whole process can be seen in Figure X.

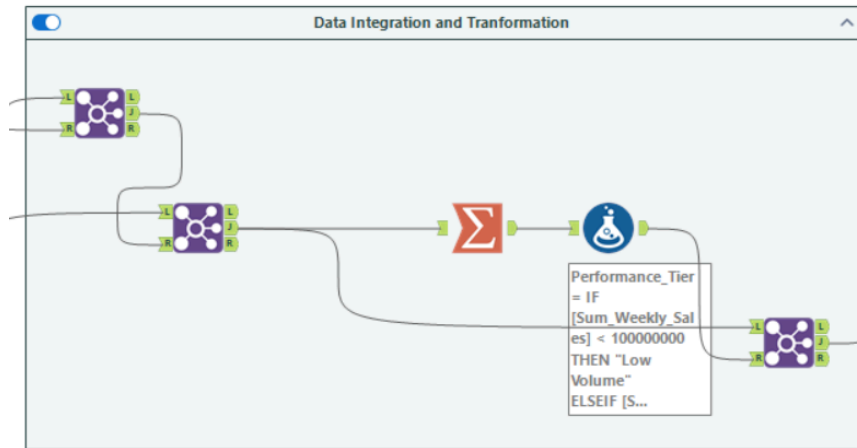


Figure 3: Data Transformation and Integration Workflow

1.4 Feature Creation

During the feature creation workflow, new analytical fields are created from the integrated sales data from the data transformation and integration phase, which might not be possible to obtain from raw data itself. Throughout the workflow, techniques like segmentation, trend decomposition, relative performance measurement and economic impact scoring are utilized. All planned analyses are supported by the enhanced dataset through each derived feature, which is aligned to the four project objectives.

- 1. Formula Tool First:** It gets temporal dimensions from the Date field and creates columns including Year, Month, Quarter, Week_Of_Year, Is_Q4, Is_Holiday_Numeric, and Dept_Store_Key. These derived fields provided the basis for structuring time and composite identifies to conduct trend and seasonal analysis of the data.
- 2. Formula Tool Second:** This was used to derive store level segmentation and performance normalization features. It splits stores into size segments and creates Store_size_segment, normalizes sales by store size (Sales_per_sqft) and ordinalizes tier labels (Perf_tier_numeric) for stores. These features enable to measure the efficiency of the sales of stores of different physical size objectively.
- 3. Formula Tool Third:** Used to manipulate the raw Temperature field to create analytical features to be used in environmental sensitivity analysis. It bucketizes the temperature readings under the Cold, Mild, and Hot category to produce Temp_band and it also takes a base temperature and produces Temp_deviation. These features enable sales patterns to be compared to the temperatures during the various time periods.
- 4. Summary Tool:** All the records are grouped together and the mean (average) value of the Weekly_sales column is calculated and the result is displayed as the value of

Avg_store_sales. This is a baseline value that is aggregated and will be re-clustered with the complete dataset to enable calculation of relative sales performance.

5. **Join Tool:** It applies the store-level average table output from the summarize tool to the original table, with the Store join key and as a LeftJoin to include all of the records in the original table. This allows the Avg_store_sales value to be carried downstream to the record level for features.
6. **Formula Tool Fourth:** Arithmetically calculates the Sales_vs_storeAvg, Holiday_sales_lift, Econ_stress_index and High_unemployment fields based on joined Avg_store_sales and economic indicator fields conditional logic. These features are directly connected to the economic conditions, periods of holiday and sales performance as quantified.
7. **Sort Tool:** It sorts the dataset so that the data is ordered by Dept in ascending, and Weekly_sales in descending order, within each Dept group. This ordering needs to be maintained in the next Multi-Row Formula tool, called sequential ranking logic.
8. **Multi-Row Formula Tool:** It returns a sequential rank for each record within its Dept group, based on the order in which it is sorted by Weekly_sales. The top of Weekly_sales record for each Dept group will have a rank 1, the next record will have a rank 2 and so on. This will provide a relative performance indicator on department level, to help with store capacity or sales analysis.

The following Figure X shows the full process of the workflow.

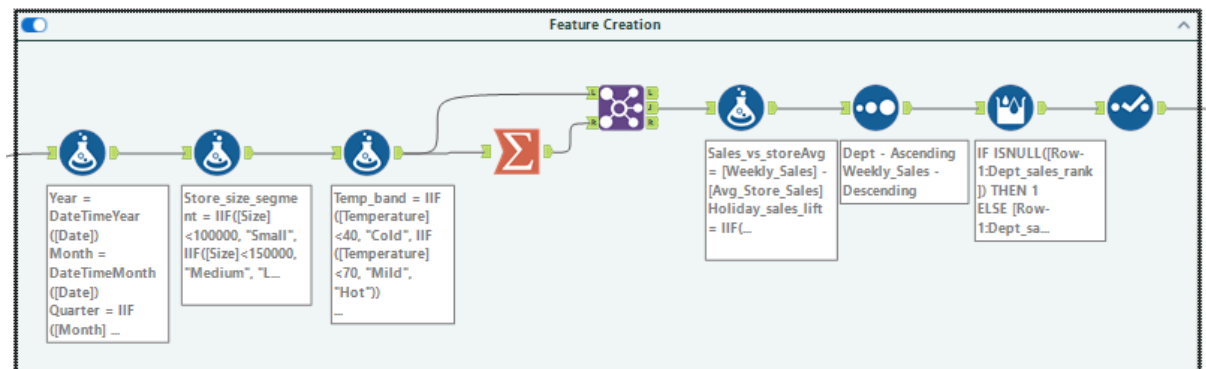


Figure 4: Feature Creation Workflow

2.0 Tools used

Alteryx Tool	Functional Purpose in This Project
Input Data	Connects to local storage to load the three raw source CSV files (feature, sales, and store) into the Alteryx workflow engine.
Browse	Used for initial data profiling to visually detect data distributions, outlier patterns, and the exact volume of missing records.
DateTime	Parses raw, text-based date strings (dd/MM/yyyy) and standardizes them into Alteryx's internal date format (yyyy-mm-dd) to enable accurate chronological matching.
Select	Converting data types (e.g., to Int64, Double, Bool), dropping redundant columns, and renaming fields.
Data Cleansing	Replacing null values in Markdown 1-5 by replacing blank entries in with 0 (indicating no discount was active).
Imputation	Calculates the mean value of populated rows to replace the missing cells in CPI and Unemployment, preventing data loss.
Unique	Checks primary key integrity across Store, Dept, and Date to ensure zero duplicate data.
Join	Merges different tables together using primary/composite keys (e.g., dual-key match on Store + Date for sales and features) to form a unified data model.
Summarize	Aggregates detailed transaction data by grouping variables (e.g., Store or Dept) and executing mathematical calculations such as Sum or Mean.
Formula	Built custom conditional statements (IF-THEN-ELSE) and expressions to develop advanced analytical dimensions (i.e. store volume tiers, temperature bands, economic stress indexes).
Sort	Prepares the stream for multi-row logic by sorting records in sequence based on parameters (sorting by Dept ascending and Weekly_Sales descending).
Multi-Row Formula	It compares data across rows within independent groups and returns a sequential performance rank (1, 2, 3...) based on sales performance.

3.0 Challenges Faced

Issue 1: The Features data set.csv contained missing values across two different types of variables in the Markdown columns and both CPI and Unemployment. Applying a general cleaning strategy or deleting rows can result in severely skewed economic analysis or massive data loss.

Resolution 1: Applied two different data cleaning strategies based on business logic. The Data Cleansing tool was used to choose the Markdown 1-5 columns and replace their Nulls with 0, based on the business rule that a blank cell indicates no promotional discount was active. At the same time, the Imputation tool was used to calculate the mean value of the populated rows for CPI and Unemployment and replace the missing cells with those values respectively.

Issue 2: Joining Sales and Features solely on Store resulted in massive, inaccurate Cartesian record multiplication.

Resolution 2: Configured a dual-key configuration match inside **Join** utilizing both Store and Date concurrently, forcing perfect historical alignment across both datasets.

Issue 3: The Week_Of_Year field caused a parse error when the Formula tool was created because the DateTimeFormat function produced a string value, while the output field was configured to be a numeric data type.

Resolution 3: The problem was overcome by putting the DateTimeFormat expression into a ToNumber function, which explicitly converts the string output to a numeric value that is compatible with the Int16 data type.